

Modeling the Temporary Impact Function $g_t(x)$

Alexander Roesler 07/31/2025

Thinking about the way limit order books behave in real life, the simple linear $g_t(x)$ example was clearly an oversimplification. Empirical temporary impact curves are flat for small orders (within the displayed spread) and then follow a concave power-law tail as order size grows. In modern equity markets, deep displayed liquidity often spans several price levels, so a piecewise model captures this:

$$g_t(x) = \begin{cases} c, & 0 \leq x \leq D_t, \\ a x^p, & x > D_t, \end{cases} \quad a = \frac{c}{D_t^p}, \quad 0 < p < 1,$$

where:

- c is the flat marginal cost (half-spread) for small trades,
- D_t is the time-varying displayed depth at the best ask,
- p governs the concavity of the tail beyond D_t .

Estimating c I computed each symbol's median half-spread during regular trading hours, then took the median of those three medians to obtain a global $c = 0.030$.

Estimating D_t For each minute of the 390-minute session, I:

1. Averaged the first non-zero ask-size per quote event within each symbol-day panel,
2. Took the cross-sectional mean across all symbols and days,
3. Fitted a penalized cubic B-spline to these 390 points to obtain a smooth function D_t .

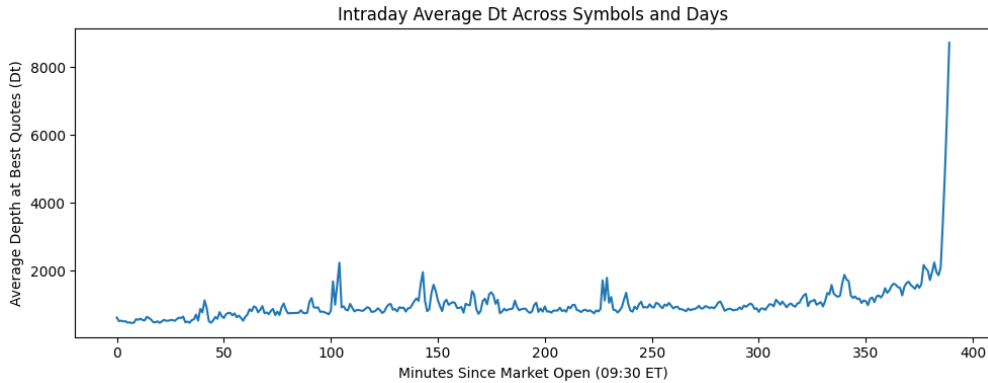


Figure 1: Intraday average D_t across symbols and days.

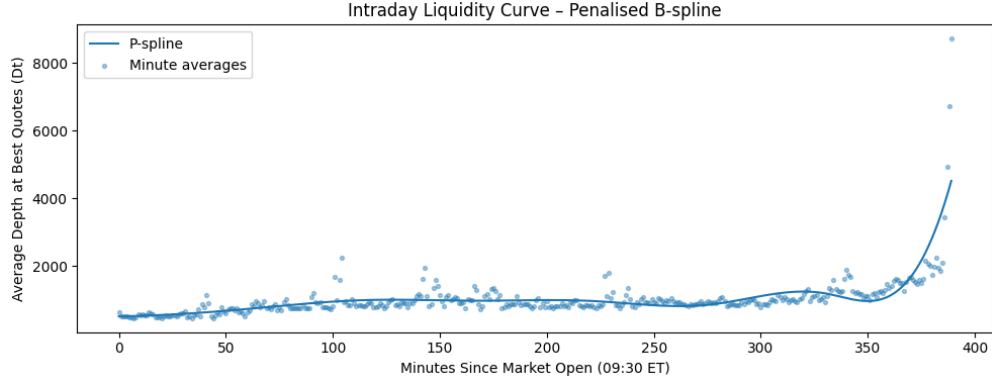


Figure 2: Penalized B-spline fit to minute-level D_t data points.

Estimating p I built an empirical impact curve by averaging, for each integer order size x , the premium $g_t(x) = \frac{\sum_{i: x_i \leq x} p_i s_i}{x}$ – best-ask across all snapshots, filtered out buckets above 1 000 shares to avoid sparse noise, and then fitted $\log g(x) = \log a + p \log x$ on the remaining averaged points.

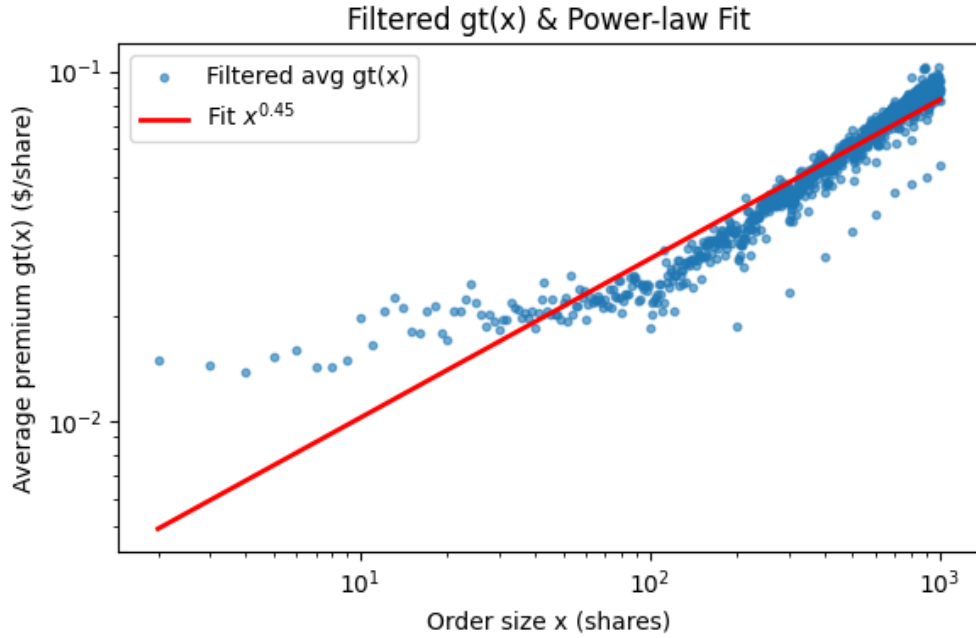


Figure 3: Filtered average $g(x)$ versus x and power-law fit $\propto x^p$.

This procedure yielded $p = 0.455$, completing the calibration of my piecewise temporary impact model $g_t(x)$.